

Harry Rogers

Test & FPGA Verification Engineer

Specializing in Hardware-in-the-Loop (HIL) • Electro-Optics Testing • FPGA Verification

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HONOR

IET Prize Winner 2026 • Awarded by The Institution of Engineering and Technology

1st Class BEng (Hons) Electronic & Computer Engineering

● EXECUTIVE SUMMARY

Accomplished **Test Engineer & FPGA Verification Specialist** with a First-Class BEng (Hons) in Electronic and Computer Engineering and recipient of the **IET Prize 2026**. Proven technical expertise spanning military-grade electro-optical surveillance system testing, hardware-in-the-loop (HIL) testbench design, and automated digital logic verification. Developer of **VAIDAR**, a novel Python-native execution engine bridging machine learning models with physical Xilinx Artix-7 FPGA silicon for accelerated coverage closure. Adept at full-lifecycle Integration, Verification & Validation (IV&V), requirement traceability, protocol analysis, and cross-functional engineering collaboration.

● CORE TECHNICAL COMPETENCIES MATRIX

FPGA & Digital Logic Design

Verilog HDL, VHDL, Xilinx Vivado Design Suite, Xilinx Artix-7, Zynq-7000, RTL Simulation (ModelSim/Vivado Simulator), Logic Synthesis, XDC Timing Constraints, Static Timing Analysis, Finite State Machines (FSM).

Hardware-in-the-Loop (HIL) & Automation

Python HIL Test Engines, PyVISA, PySerial, PyTorch (AI-Driven Stimulus Generation), Automated Test Equipment (ATE), FAT/SAT Test Suites, Automated Telemetry & Failure Triage.

Electro-Optics & System Testing

Electro-Optical Surveillance Payloads, Thermal & Daylight Cameras, Servo Gimbals, Laser Rangefinders, Acceptance Test Procedures (ATP), DOORS Traceability, Fault Diagnostics.

Protocols & Lab Instrumentation

Oscilloscopes, Logic Analyzers, Spectrum Analyzers, JTAG Boundary Scan, UART, PCIe, Ethernet, SPI, I2C, CAN Bus, RS-422/485, CoaXPress, Camera Link, SDI.

● PROFESSIONAL WORK EXPERIENCE

Test Engineer

Chess Dynamics • Sussex, UK

2024 - Present

Specialist defence & security enterprise manufacturing ultra-precise electro-optical tracking, surveillance, and gimbals.

- System-Level Verification & Testing:** Leads comprehensive testing and commissioning for multi-sensor electro-optical surveillance heads, pan-tilt gimbals, thermal imagers, and radar tracking assemblies.
- Test Procedure Authoring:** Authors and executes formal Acceptance Test Procedures (ATP), Factory Acceptance Tests (FAT), and Site Acceptance Tests (SAT) in compliance with stringent military and industrial quality standards.
- Automated Test Suite Development:** Designs Python and hardware-based automated scripts for sensor calibration, optical target collation, and continuous regression testing, reducing manual test setup time by 40%.
- Cross-Functional Fault Isolation:** Isolates complex root-cause anomalies across multi-board embedded microprocessors, video transmission protocols (Camera Link/SDI), and servo control feedback loops.
- Instrumentation Mastery:** Operates precision optical collimators, thermal blackbody targets, laser power meters, logic analyzers, and high-bandwidth digital storage oscilloscopes.

Systems Engineer — Integration, Verification & Validation (IV&V)

12-Month Industrial Placement

BAE Systems • UK

Global defence, aerospace, and security contractor developing cutting-edge mission systems.

- IV&V Lifecycle Execution:** Executed end-to-end Integration, Verification & Validation (IV&V) protocols for mission-critical hardware and electronic system architectures.
- HIL Testbench Assembly & Rig Operation:** Constructed, wired, and calibrated hardware-in-the-loop (HIL) testbenches, establishing real-time signal injection and telemetry feedback loops between physical hardware and host simulators.
- Requirements Verification Traceability:** Managed requirements verification matrices within IBM DOORS, mapping high-level system requirements to granular hardware/firmware test cases with 100% audit compliance.
- Protocol & Signal Integrity Analysis:** Debugged multi-channel bus communication protocols including MIL-STD-1553, ARINC 429, RS-422, and Ethernet using protocol analyzers and logic state analyzers.
- Defect Logging & Verification Reports:** Documented technical non-conformance reports, root-cause assessments, and formal verification summaries presented during engineering review boards (ERBs).

FEATURED ENGINEERING & RESEARCH PROJECTS

VAIDAR Framework

Verification & AI for Digital Architecture Runtime

Final Year Project • Passed with 1st Distinction

Designed and built a novel Python-native Hardware-in-the-Loop (HIL) execution engine for Xilinx Artix-7 FPGAs to solve the industry bottleneck where hardware verification consumes up to 70% of the FPGA development lifecycle.

Silicon-to-Software Bridge

Establishes bi-directional Python-to-FPGA physical serial/PCIe transport for real-time vector streaming.

AI-Driven Stimulus Engine

Integrates PyTorch models to generate adaptive stimulus vectors targeting unvisited state-machine branches.

- Achieved 99.9%+ logic coverage closure in 65% less runtime compared to standard random test vectors.
- Built an automated NLP/clustering log triage engine that automatically groups regression failures by root-cause signature.

Automated Electro-Optical Target Collimation Test Bench

Chess Dynamics & Applied Optics

Engineered an automated script-driven calibration routine for multi-spectral optical payloads. Synchronized precision pan-tilt stages with infrared blackbody emitters to measure Minimum Resolvable Temperature Difference (MRTD) and optical axis bore-sighting, delivering repeatable metrics and eliminating operator bias.

High-Speed FPV Avionics & Telemetry Systems

Independent Systems Engineering

Designed, built, and flight-tested custom FPV quadcopter systems featuring low-latency analog/digital video transmission, custom RF link tuning, and real-time blackbox flight telemetry logging. Implemented PID loop optimization and ESC motor timing for high-dynamic maneuvers.

● EDUCATION & ACADEMIC DISTINCTION

BEng (Hons) Electronic & Computer Engineering

University of Brighton • First-Class Honours (1st Class)

Graduated 2026

Key Curriculum Modules

- Digital Logic Design & Synthesis (Verilog / Vivado)
- Embedded Microprocessor Architectures (C / Assembly)
- Hardware Verification & Testing Methodologies
- Digital Signal Processing (DSP) & Control Systems
- RF Communications & High-Speed Data Links

Official Credentials

Degree Result: 1st Class Honours (First-Class)

Official PDF Link: [Brighton Degree Certificate PDF ↗](#)

Final Year Project (XE636): Distinction Grade (VAIDAR HIL Framework)

● AWARDS & PROFESSIONAL DISTINCTIONS

WINNER The IET Prize 2026

Awarded by **The Institution of Engineering and Technology (IET)** for outstanding distinction and academic excellence in Electronic & Computer Engineering.

Signatories: Ed Almond (Chief Executive), Dawn Ohlson (IET President 2025-26), Warren East (Board of Trustees).

BAE Systems IV&V Performance Commendation

Recognized during industrial placement for exceptional setup of hardware-in-the-loop testbench rigs and zero-defect requirements verification execution.

BAE Systems

● COMPREHENSIVE TOOLING & TECHNICAL INDEX

HDL & EDA Tools

- Verilog HDL
- VHDL
- Xilinx Vivado
- ModelSim
- Vivado Simulator
- XDC Constraints

Software & AI

- Python 3.x
- PyTorch
- NumPy / SciPy
- PyVISA / Serial
- C / C++
- MATLAB / Simulink

Bus Protocols

- UART / RS-422
- PCIe
- Ethernet
- SPI / I2C
- CAN Bus
- MIL-STD-1553

Lab & Systems

- Oscilloscopes
- Logic Analyzers
- Spectrum Analyzers
- JTAG Boundary Scan
- IBM DOORS
- Git & CI/CD

● PROFESSIONAL AFFILIATIONS & REFERENCES

Professional Membership: Member of The Institution of Engineering and Technology (IET).

References: Professional and academic references from Chess Dynamics, BAE Systems, and the University of Brighton are available upon request.